Banking Transaction Processing

Latar Belakang

Sebuah sistem perbankan ingin memproses transaksi transfer antar rekening secara asynchronous.

Namun sistem memiliki aturan:

1. transaksi maksimal diproses selama **4 detik**. jika lebih transaksi dianggap timeout.
2. karena komunikasi antar bank, proses transaksi memiliki waktu random **1-5 detik**.

Saat transaksi tidak melebihi 4 detik:

1. Sistem melakukan validasi saldo pengirim.
2. Memastikan konsistensi data saldo.

Jika saldo tidak mencukupi maka transaksi gagal dan tampilkan informasi:

1. ID transaksi
2. rekening asal
3. rekening tujuan
4. nominal transfer
5. saldo saat ini

Jika saldo mencukupi, tampilkan informasi:

1. saldo dan no rekening pengirim
2. saldo dan no rekening penerima

Sample Data:

Bank	No Rekening	Saldo
CIMB	C001	Rp 300000
MANDIRI	M002	Rp 500000
BNI	B003	Rp 400000
BCA	BC04	Rp 800000

ID TRX	From Rekening	To Rekening	Amount
1	C001	M002	Rp 300000
2	C001	B003	Rp 600000
3	M002	BC04	Rp 200000
4	B003	C001	Rp 500000
5	BC04	M002	Rp 700000
6	C001	M002	Rp 400000
7	B003	C001	Rp 300000

Sample Result:

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15:01:40 | Processing TX-6 | C001 -> M002 | Rp400000
15:01:40 | Processing TX-1 | C001 -> M002 | Rp300000
15:01:40 | Processing TX-5 | BC04 -> M002 | Rp700000
15:01:40 | Processing TX-2 | C001 -> B003 | Rp600000
15:01:40 | Processing TX-3 | M002 -> BC04 | Rp200000
15:01:40 | Processing TX-4 | B003 -> C001 | Rp500000
15:01:40 | Processing TX-7 | B003 -> C001 | Rp300000
15:01:41 | TX-5 SUCCESS
   Saldo BC04 sekarang Rp100000
   Saldo M002 sekarang Rp1200000
15:01:41 | TX-7 SUCCESS
   Saldo B003 sekarang Rp100000
   Saldo C001 sekarang Rp600000
15:01:41 | TX-1 SUCCESS
   Saldo C001 sekarang Rp300000
   Saldo M002 sekarang Rp1500000
15:01:42 | TX-4 FAILED | B003 -> C001 | Transfer Rp500000 | Saldo B003 hanya Rp100000
15:01:43 | TX-6 FAILED | C001 -> M002 | Transfer Rp400000 | Saldo C001 hanya Rp300000
15:01:44 | TX-3 SUCCESS
   Saldo M002 sekarang Rp1300000
   Saldo BC04 sekarang Rp300000
15:01:45 | TX-2 TIMEOUT

===== FINAL BALANCE =====
CIMB (C001) : Rp300000
MANDIRI (M002) : Rp1300000
BNI (B003) : Rp100000
BCA (BC04) : Rp300000

Process finished with the exit code 0
```

```
15:13:55 | Processing TX-7 | B003 -> C001 | Rp300000
15:13:55 | Processing TX-2 | C001 -> B003 | Rp600000
15:13:55 | Processing TX-1 | C001 -> M002 | Rp300000
15:13:55 | Processing TX-3 | M002 -> BC04 | Rp200000
15:13:55 | Processing TX-6 | C001 -> M002 | Rp400000
15:13:55 | Processing TX-4 | B003 -> C001 | Rp500000
15:13:55 | Processing TX-5 | BC04 -> M002 | Rp700000
15:13:56 | TX-6 FAILED | C001 -> M002 | Transfer Rp400000 | Saldo C001 hanya Rp300000
15:13:57 | TX-1 SUCCESS
   Saldo C001 sekarang Rp0
   Saldo M002 sekarang Rp800000
15:13:57 | TX-4 FAILED | B003 -> C001 | Transfer Rp500000 | Saldo B003 hanya Rp400000
15:13:58 | TX-5 SUCCESS
   Saldo BC04 sekarang Rp100000
   Saldo M002 sekarang Rp1500000
15:13:58 | TX-2 FAILED | C001 -> B003 | Transfer Rp600000 | Saldo C001 hanya Rp0
15:13:59 | TX-7 SUCCESS
   Saldo B003 sekarang Rp100000
   Saldo C001 sekarang Rp300000
15:14:00 | TX-3 TIMEOUT

===== FINAL BALANCE =====
CIMB (C001) : Rp300000
MANDIRI (M002) : Rp1500000
BNI (B003) : Rp100000
BCA (BC04) : Rp100000

Process finished with the exit code 0
```

```
15:12:25 | Processing TX-7 | B003 -> C001 | Rp300000
15:12:25 | Processing TX-3 | M002 -> BC04 | Rp200000
15:12:25 | Processing TX-6 | C001 -> M002 | Rp400000
15:12:25 | Processing TX-1 | C001 -> M002 | Rp300000
15:12:25 | Processing TX-5 | BC04 -> M002 | Rp700000
15:12:25 | Processing TX-2 | C001 -> B003 | Rp600000
15:12:25 | Processing TX-4 | B003 -> C001 | Rp500000
15:12:26 | TX-7 SUCCESS
   Saldo B003 sekarang Rp100000
   Saldo C001 sekarang Rp600000
15:12:26 | TX-5 SUCCESS
   Saldo BC04 sekarang Rp100000
   Saldo M002 sekarang Rp1200000
15:12:26 | TX-3 SUCCESS
   Saldo M002 sekarang Rp1000000
   Saldo BC04 sekarang Rp300000
15:12:27 | TX-2 SUCCESS
   Saldo C001 sekarang Rp0
   Saldo B003 sekarang Rp700000
15:12:28 | TX-4 SUCCESS
   Saldo B003 sekarang Rp200000
   Saldo C001 sekarang Rp500000
15:12:28 | TX-1 SUCCESS
   Saldo C001 sekarang Rp200000
   Saldo M002 sekarang Rp1300000
15:12:30 | TX-6 TIMEOUT

===== FINAL BALANCE =====
CIMB (C001) : Rp200000
MANDIRI (M002) : Rp1300000
BNI (B003) : Rp200000
BCA (BC04) : Rp300000

Process finished with the exit code 0
```

```
15:14:37 | Processing TX-4 | B003 -> C001 | Rp500000
15:14:37 | Processing TX-5 | BC04 -> M002 | Rp700000
15:14:37 | Processing TX-1 | C001 -> M002 | Rp300000
15:14:37 | Processing TX-7 | B003 -> C001 | Rp300000
15:14:37 | Processing TX-2 | C001 -> B003 | Rp600000
15:14:37 | Processing TX-6 | C001 -> M002 | Rp400000
15:14:37 | Processing TX-3 | M002 -> BC04 | Rp200000
15:14:38 | TX-3 SUCCESS
   Saldo M002 sekarang Rp300000
   Saldo BC04 sekarang Rp1000000
15:14:38 | TX-2 FAILED | C001 -> B003 | Transfer Rp600000 | Saldo C001 hanya Rp300000
15:14:38 | TX-5 SUCCESS
   Saldo BC04 sekarang Rp300000
   Saldo M002 sekarang Rp1000000
15:14:39 | TX-4 FAILED | B003 -> C001 | Transfer Rp500000 | Saldo B003 hanya Rp400000
15:14:39 | TX-6 FAILED | C001 -> M002 | Transfer Rp400000 | Saldo C001 hanya Rp300000
15:14:39 | TX-7 SUCCESS
   Saldo B003 sekarang Rp100000
   Saldo C001 sekarang Rp600000
15:14:41 | TX-1 SUCCESS
   Saldo C001 sekarang Rp300000
   Saldo M002 sekarang Rp1300000

===== FINAL BALANCE =====
CIMB (C001) : Rp300000
MANDIRI (M002) : Rp1300000
BNI (B003) : Rp100000
BCA (BC04) : Rp300000

Process finished with the exit code 0
```